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Predictive Maintenance

Bearing Remaining Useful Life Prediction

FEMTO / PRONOSTIA Dataset · IEEE PHM 2012 Challenge

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Industrial AI — Mini Project 2

Business Problem

Ball bearing failures are the #1 cause of unplanned downtime on production lines with heavily loaded electric motors. Each failure means production stops, emergency repair costs, and missed deliveries.

Q1 — How long?

How many hours until this bearing needs replacing?

→ RUL Regression

Q2 — What stage?

Is it healthy, degrading, or about to fail?

→ 4-Class Classification

Q3 — How sure?

How confident are we in that estimate?

→ ±MAE Uncertainty Band

All three outputs combined → Production engineer can make a maintenance decision without a data scientist.

Dataset & Methodology

FEMTO / PRONOSTIA Dataset

6	Run-to-failure bearings (training)
3	Operating conditions (1800/1650/1500 rpm)
5,897	Total 0.1s vibration windows
2,560	Samples per window @ 25.6 kHz
10s	Interval between windows
1.4h – 7.8h	Bearing lifetime range

CRISP-DM Workflow

01	**Business Understanding**	Define the 3 engineering questions
02	**Data Understanding**	EDA — raw signals, RMS trends, lifetime analysis
03	**Data Preparation**	Bandpass filter ⟶ 10 features ⟶ HI ⟶ RUL labels
04	**Modelling**	CNN + SVR (regression) + RF (classification)
05	**Evaluation**	LOBO validation — leave one bearing out
06	**Deployment**	Maintenance dashboard with uncertainty bands

Signal structure: every 10 seconds, a 0.1s burst is recorded at 25.6 kHz ⟶ 2560 samples/window. Temperature excluded (inconsistent across bearings). Torque not present in dataset.

Feature Engineering: What We Extract from Vibration

Bandpass filter: 2–10 kHz → Isolates bearing fault frequencies. Below 2 kHz = shaft noise. Above 10 kHz = unrelated high-freq noise. (Sutrisno 2012, Cheng 2018)

RMS

Primary degradation indicator — rises steadily as bearing surface wears. Most important feature (23% RF importance).

Kurtosis

Measures signal spikiness. Healthy ≈ 3. Rises early when fault impulses appear — catches faults before RMS changes.

Crest Factor

Moves OPPOSITE to RMS as fault progresses. Together RMS↑ + Crest↓ = confirmed advanced degradation.

Skewness

Healthy ≈ 0. Shifts when fault asymmetry appears. Independent information from kurtosis.

Impulse Factor

More sensitive to single sharp spikes than crest factor. Top-3 feature in SHAP analysis on FEMTO.

Each feature computed for H and V channels → 10 features total per window

Health Indicator: Tracking Degradation 0 ⟶ 1

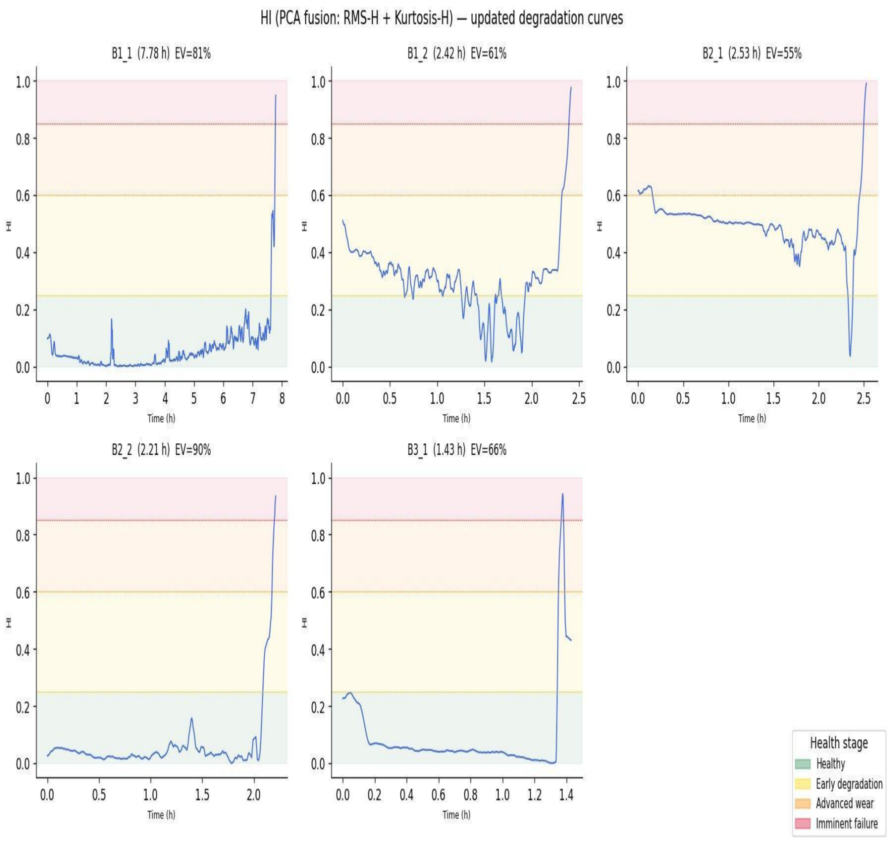
Why we need an HI

Different bearings have wildly different vibration levels and lifetimes (1.4h to 7.8h). Raw features are incomparable across conditions. We need a single normalised score that means the same thing for every bearing.

Construction — PCA fusion

1. Smooth both RMS and kurtosis trends
2. Combine them using PCA: finds the shared degradation direction
3. Normalise to 0→1 — flip if going wrong direction

PC1 explained variance across bearings: 55% – 90%
RUL-based classes (25/35/25/15%) — consistent across all conditions



Modelling — CNN + SVR + Random Forest

1D CNN — Feature Extractor

Reads raw vibration signal directly — no manual feature selection

Three layers of pattern detection — from broad energy changes to subtle fault signatures

Compresses what it learns into a fixed summary per window

Trained to predict how much life the bearing has left

SVR — RUL Regression

Takes CNN summary + our 10 hand-crafted features as input

Predicts hours remaining, normalised per bearing so lifetimes are comparable

Predictions smoothed over 2.5 minutes: RUL can't jump between 10-second windows

Output: RUL in hours with ±MAE confidence band

Literature ref: Cheng et al. 2018 (validated on FEMTO)

Random Forest — Health Stage Classification

Input: 10 hand-crafted features (X matrix)

200 estimators   class_weight=balanced

Output: 4 health stages — Healthy / Early / Advanced / Imminent

Why LOBO Validation?

Leave-One-Bearing-Out: train on 4 bearings, test on the unseen 1. Repeats 5×. Standard in FEMTO literature: measures true generalisation to a bearing the model has never seen, matching real deployment conditions.

RUL Regression Results: LOBO Validation

Bearing	R^2 (baseline)	R^2 (smoothed)	MAE (hours)	Note
B1_1	-0.475	-0.608	2.29h	Lifetime outlier (7.78h vs avg 2.3h)
B1_2	0.529	0.154	0.56h	
B2_1	0.475	0.528	0.41h	
B2_2	0.837	0.877	0.16h	Best fold
B3_1	-0.244	0.101	0.28h	Condition 3 — only 1 training bearing
Mean (all)	**0.224**	**0.210**	**0.74h**	
Mean excl. B1_1	**0.399**	**0.415**	**0.35h**	**Comparable to literature (0.3–0.7)**

B1_1 — Why negative R^2?

7.78h lifetime vs 1.4–2.5h for all others. Model trains on short bearings and must predict a 3× longer life. Known cross-condition generalisation challenge in FEMTO literature.

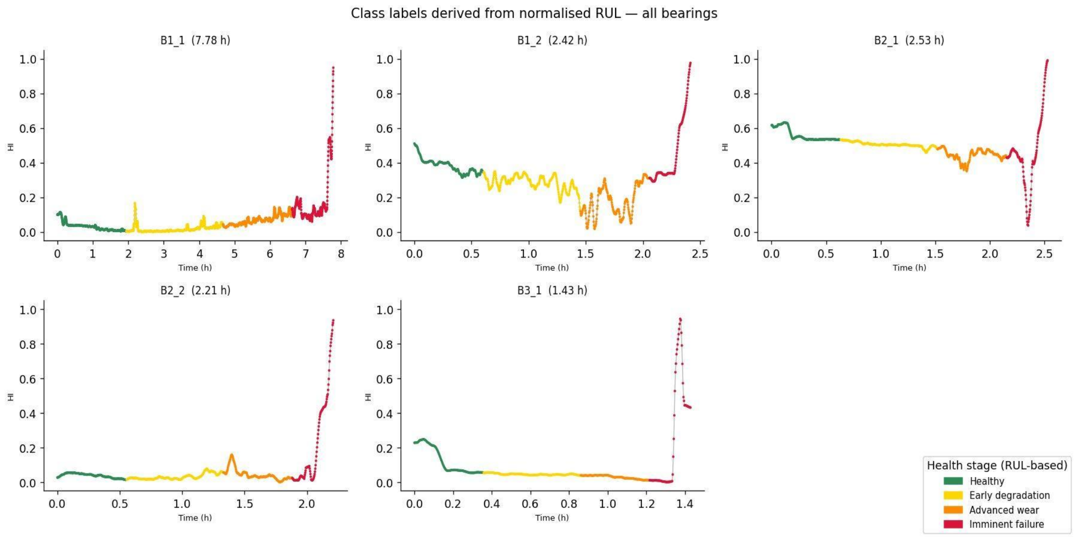
B2_2 — R^2=0.877

Best fold. Similar lifetime and condition to training set. Predicted RUL tracks the actual diagonal cleanly.

Business context

Mean MAE=0.35h (excl. B1_1) means ±21 min error on a 2h bearing life. Sufficient for shift-change maintenance scheduling.

Classification Results: 4 Health Stages



LOBO Classification Summary

Mean Accuracy	**42.1%**
Mean F1 (macro)	**36.3%**
Baseline F1	**13.0%**
RF improvement	**ΔF1 +0.234**
Best fold (B2_2)	**F1=0.547**

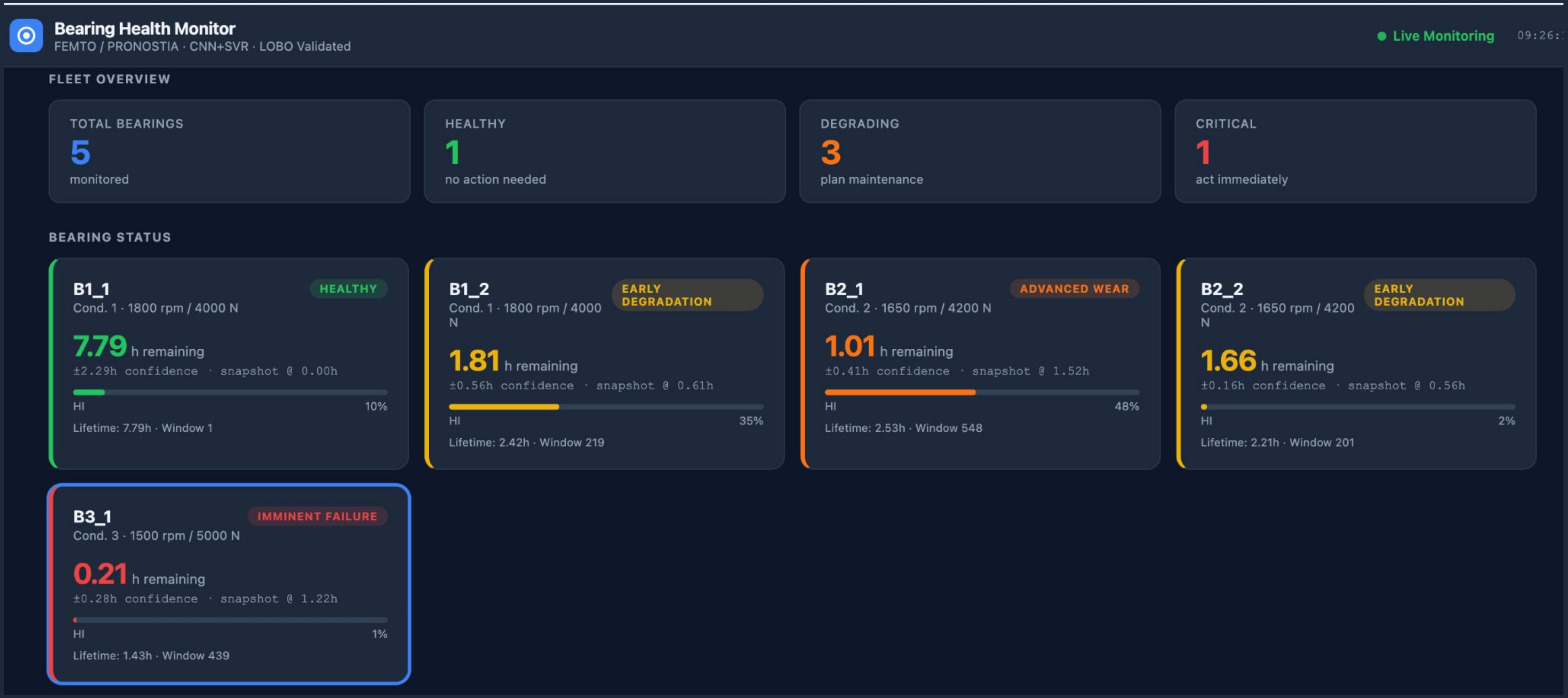
Feature Importance (Random Forest — averaged across LOBO folds)

rms_h = 23.2% · rms_v = 21.6% · kurt_h = 13.6% · kurt_v = 7.9% · impulse_h = 7.5% · crest_h = 6.3%
→ RMS dominates, confirming literature (Sutrisno 2012). Kurtosis 3rd — validates early-fault sensitivity argument.

Healthy: >75% RUL   Early degradation: 40–75%   Advanced wear: 15–40%   Imminent failure: <15%

Predictive Maintenance Dashboard

Answers the business questions — readable by a production engineer



Conclusions, Limitations & Future Work

What We Achieved

✓ End-to-end CRISP-DM pipeline on real run-to-failure data

✓ CNN + SVR: mean R^2=0.42 excl. B1_1 — comparable to literature

✓ RF classifier: F1 macro 0.363, ΔF1 +0.234 vs majority baseline

✓ PCA-fused HI gives gradual degradation visualisation

✓ Business dashboard answers all 3 engineering questions

✓ LOBO validation: honest generalisation to unseen bearings

Limitations

⚠️ B1_1 ran for 7.78h while all others failed within 1.4–2.5h: model had never seen a bearing with that much life left

⚠️ Only 5 bearings used for training: not enough to learn reliably across different speeds and loads

⚠️ Dashboard shows bearings at end of life: real deployment would alert hours before this point

⚠️ Only one bearing from the highest-load condition: not enough to learn its pattern reliably

Future Work

1. More data, same rigour: use all 17 FEMTO bearings while keeping LOBO validation
2. Richer signal features: add frequency band energy and spectral entropy
3. Better uncertainty: model that learns confidence per prediction, not a fixed ±MAE
4. Smarter health curve: Enforce degradation score can only go up over time

References: [1] Cheng, Ma et al. 2018 — CNN+SVR on FEMTO [2] Sutrisno, Oh et al. 2012 — Kurtosis + Bayesian Monte Carlo [3] PRONOSTIA dataset, FEMTO-ST Institute, IEEE PHM 2012

Thank You!